Computer Vision Laboratory – IX

Experiment 1: Visualization of CNN Filters and Feature Maps

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Abstract

This experiment focuses on understanding the internal behavior of Convolutional Neural Networks (CNNs) through visualization techniques. CNNs are widely used for image classification, but their decision-making process is often difficult to interpret. To address this, the experiment explores the visualization of convolutional filters and feature maps, which provide insights into how the network processes images.

A CNN model was trained on the CIFAR-10 dataset, and the learned filters from convolutional layers were extracted and visualized to observe the patterns captured by the network. Additionally, feature maps were generated by passing an input image through the trained model, allowing visualization of how different layers transform the image representation.

The results demonstrate that early layers capture basic features such as edges and textures, while deeper layers capture more complex patterns and object-level information. This experiment highlights the importance of visualization in improving the interpretability and understanding of deep learning models.

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1. Aim

To visualize and understand the internal workings of a **Convolutional Neural Network (CNN)** by analyzing learned **filters** and generated **feature maps**.

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2. Objective

- To understand how CNNs extract features from images
- To visualize convolutional filters learned during training
- To visualize feature maps generated at different layers
- To interpret how CNN processes images layer by layer
- To implement visualization techniques using TensorFlow

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3. Introduction

Convolutional Neural Networks (CNNs) are widely used for image classification tasks due to their ability to automatically learn hierarchical features from images. However, CNNs are often considered "black-box" models because their internal operations are not directly interpretable.

To better understand CNN behavior, visualization techniques are used. These techniques help in analyzing:

- **Filters (kernels):** Learned weights that detect patterns such as edges and textures
- **Feature Maps:** Outputs of convolution layers that highlight important regions in an image

By visualizing filters and feature maps, we can gain insight into how CNNs progressively transform input images into meaningful representations.

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4. Dataset Description

The experiment uses the **CIFAR-10 dataset**, which contains:

- **60,000 color images**
- **10 classes**
- **Image size: 32 × 32 pixels**

Classes:

- Airplane
- Automobile
- Bird
- Cat

- Deer
- Dog
- Frog
- Horse
- Ship
- Truck

Dataset Split:

Dataset	Number of Images
Training	50,000
Testing	10,000

5. Theory

5.1 Convolutional Filters

Filters (or kernels) are small matrices of weights that slide over the input image. Each filter detects specific patterns such as:

- Edges
- Corners
- Textures

During training, the network learns optimal filter values.

5.2 Feature Maps

Feature maps are the outputs obtained after applying convolution operations.

- Each filter produces one feature map
- Feature maps highlight important regions of the image
- Deeper layers capture more complex patterns

5.3 Hierarchical Feature Learning

CNNs learn features in stages:

Layer Type	Features Learned
Early Layers	Edges, lines
Middle Layers	Textures, shapes
Deep Layers	Objects, patterns

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6. Methodology

The following steps were performed:

Step 1: Dataset Loading

- CIFAR-10 dataset was loaded

Step 2: Preprocessing

- Images were normalized
- Labels were converted to categorical format

Step 3: CNN Model Construction

- Built a CNN with multiple convolution and pooling layers

Step 4: Model Training

- Model was trained using training dataset

Step 5: Filter Visualization

- Extracted weights from convolution layers
- Plotted filters using visualization techniques

Step 6: Feature Map Visualization

- Created intermediate model
- Passed input image through network
- Visualized outputs of convolution layers

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7. Model Architecture

- Conv2D (32 filters) → ReLU → MaxPooling
- Conv2D (64 filters) → ReLU → MaxPooling
- Conv2D (128 filters) → ReLU

- Flatten
- Dense (128) → ReLU
- Dense (10) → Softmax

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8. Training Parameters

Parameter	Value
Optimizer	Adam
Loss Function	Categorical Crossentropy
Epochs	5
Batch Size	64
Metric	Accuracy

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9. Results

9.1 Model Performance

Metric	Value
Training Accuracy	74.73599910736084
Test Accuracy	69.19000148773193

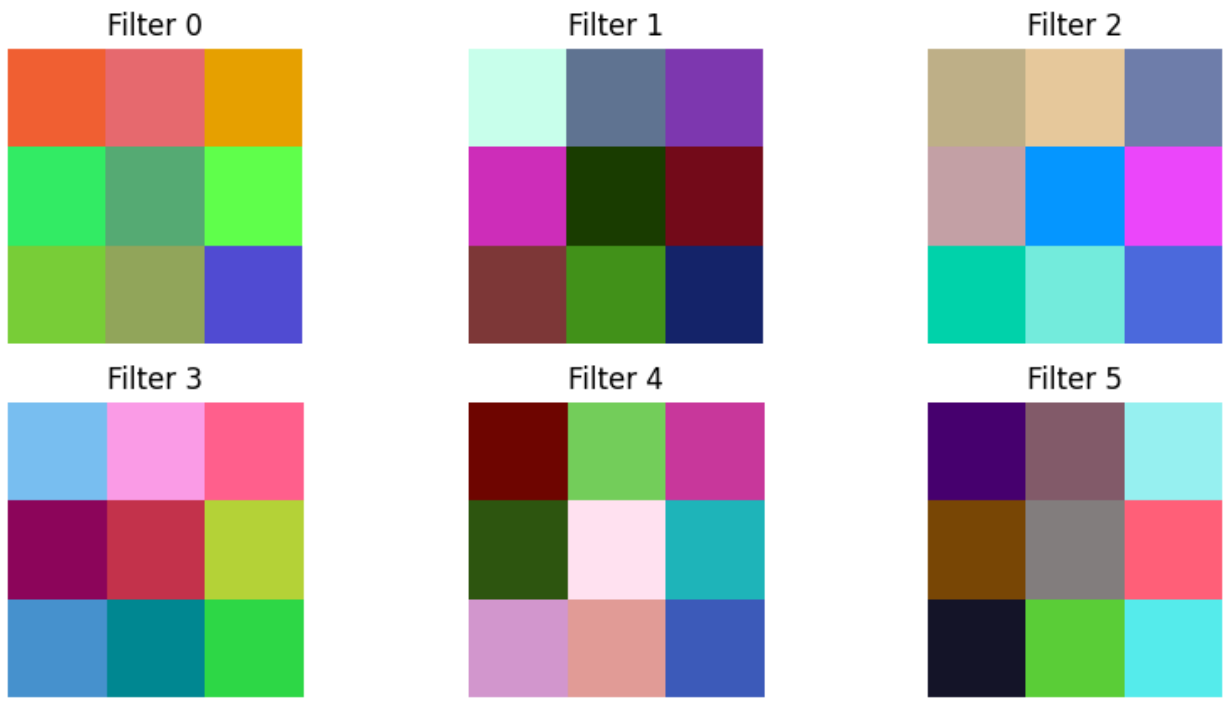
```
train_loss, train_acc = model.evaluate(x_train, y_train, verbose=0)
test_loss, test_acc = model.evaluate(x_test, y_test, verbose=0)

print("Training Accuracy:", train_acc * 100)
print("Test Accuracy:", test_acc * 100)
```

```
Training Accuracy: 74.73599910736084
Test Accuracy: 69.19000148773193
```

9.2 Filter Visualization

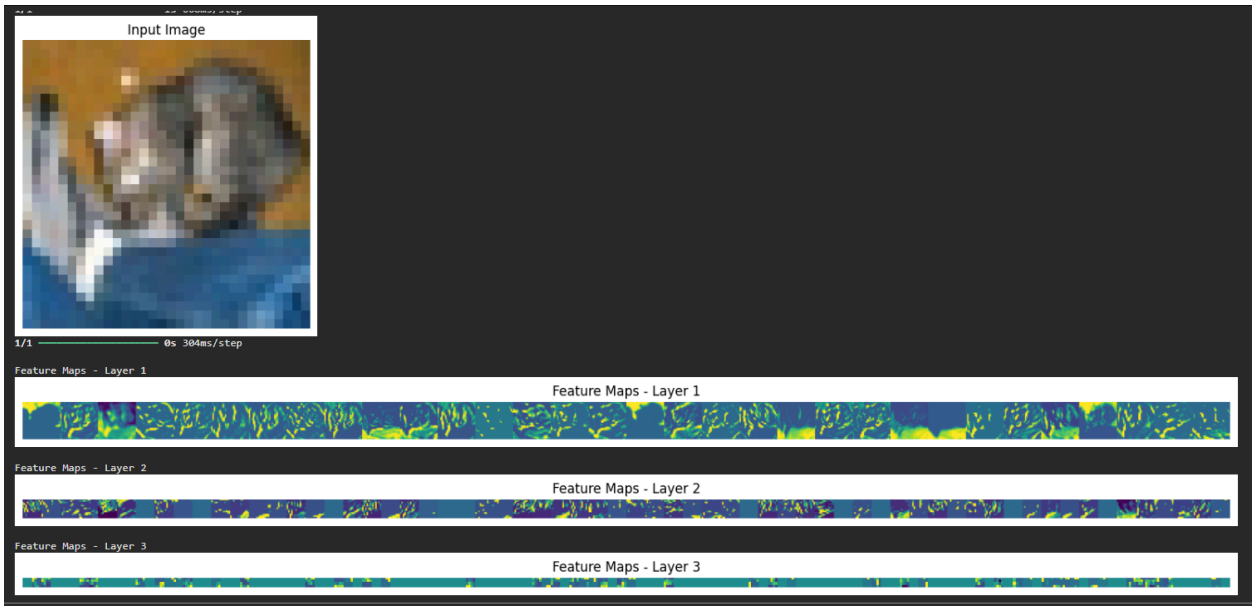
Filters from the first convolution layer were visualized. These filters capture basic features such as edges and color gradients.



9.3 Feature Map Visualization

Feature maps were generated for different convolution layers. Observations:

- Early layers highlight edges and textures
- Deeper layers highlight complex patterns



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10. Observations

- CNN automatically learns important features
- Filters act as pattern detectors
- Feature maps show how image representation evolves
- Deeper layers capture more abstract information

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11. Applications

- Image classification
- Object detection
- Medical image analysis
- Face recognition
- Autonomous vehicles

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12. Advantages

- Helps interpret CNN behavior
- Improves understanding of deep learning models
- Useful for debugging and improving models

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13. Limitations

- Visualization can be complex
- Hard to interpret deeper layers
- Requires computational resources

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14. Conclusion

In this experiment, the internal functioning of a CNN was explored through visualization of filters and feature maps. It was observed that CNN layers progressively extract meaningful features from images, starting from simple edges to complex patterns.

These visualization techniques provide valuable insights into how deep learning models process visual data and help improve model interpretability.

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15. References

1. Machine Learning Mastery – CNN Visualization
2. TensorFlow Documentation
3. Deep Learning – Ian Goodfellow